

## RS004 Week 12 — English Worksheet

**Topic:** Coin Types + Spin Animation (Apply) · **Course:** Platformer Game · **Time:** about 45 minutes

This week coins come in **kinds** — gold (1 point), gem (5), secret (10). You store each coin as a **dictionary** with its rect, kind, and value, and you make them **spin** with a **sin** wave that wobbles their width.

Keep these words handy: **dictionary**, **kind / value**, **math.sin**, **wobble (animate)**, **frame**.

### 1 · Predict 🧠

Read each snippet. Before you run it, write what you think will happen.

```
coins = [
    {"rect": pygame.Rect(180, 480, 20, 20), "kind": "gold", "value": 1},
    {"rect": pygame.Rect(580, 480, 20, 20), "kind": "gem", "value": 5},
]
```

Each coin is now a dictionary. How do you read the second coin's value?

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```
wave = math.sin(frame * 0.1) * 8
w = max(2, int(20 + wave))
```

**math.sin(...)** swings between -1 and +1. What is the smallest the width **w** can get, and why the **max(2, ...)** ?

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```
if player.colliderect(coin["rect"]):  
    score = score + coin["value"]
```

**The player grabs a gem. How much does the score go up?**

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## 2 · Read the Dictionary

A coin is `{"rect": pygame.Rect(450, 250, 20, 20), "kind": "secret", "value": 10}`. Fill in the blanks.

```
coin["kind"]    = ____  
coin["value"]   = ____  
coin["rect"].x  = ____
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**Why is a dictionary better than three separate lists (one for rects, one for kinds, one for values)?**

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### 3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

**Bug A** — Every coin should add its own value, but all coins add 1 point no matter the kind.

```
for coin in coins[:]:
    if player.colliderect(coin["rect"]):
        score = score + 1
        coins.remove(coin)
```

**Hint:** use `coin["value"]`, not a fixed `1`.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — The collision check should compare the player with the coin's rectangle, but it crashes saying a dict can't collide.

```
for coin in coins[:]:
    if player.colliderect(coin):
        score += coin["value"]
```

**Hint:** the rect lives inside the dictionary under a key.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — The coins should keep wobbling, but they spin once and then freeze.

```
def draw_coin(coin, frame):  
    wave = math.sin(frame * 0.1) * 8  
    w = max(2, int(20 + wave))  
    ...  
# in the game loop:  
frame = 0  
draw_coin(coin, frame)
```

**Hint:** `frame` never increases, so `sin` never changes. Where should `frame` grow?

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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## 4 · Add a Kind

Start from your working coin types.

1. Add a fourth coin kind (for example a time bonus). Write its dictionary entry.

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2. Reposition the secret coin so it fits *your* level. Where did you place it?

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3. **Stretch:** make the spin faster for gems than for gold. Which number controls the speed?

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## 5 · Build & Show 📷

Put three kinds of spinning coins into your own level — gold, gem, and a secret coin reachable only by a tricky jump.

When it works, send a **photo or video** collecting each kind, then explain what you did. Use these starters — write 4 to 6 sentences.

First, I stored each coin as a dictionary so that ...

The different kinds add different points because ...

The spin works with `math.sin` by ...

The `max(2, ...)` stops the coin from ...

One tricky moment was ...

If I had more time, I would ...

## 6 · Record Your Walkthrough 🎥

Take a video collecting all three kinds. Teach it to someone new. Try to use these words: **dictionary**, **kind**, **value**, **sin**, **spin**.

1. Show your gold, gem, and secret coins.
2. Collect each one and show the score jump by different amounts.
3. Read your `math.sin` line out loud and explain the wobble.
4. Show the secret coin and the jump to reach it.

**Write what you will say in your video.** Plan it here first.

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Send this worksheet + your walkthrough video to teacher on KakaoTalk.